



Software Requirement Specification(SRS)

Project Title : Neighborhood Skill Exchange

Course Name : Software Engineering Laboratory

Course Code: CSE 3422

Team Console

Pratoy Barua	0112330398
Waled Rahman	0112330400
Asifur Rahman	0112330483
Md. Rejwan Hossan Asif	0112330482

Submitted To:

Rejwan Ahmed
Lecturer, Dept. of CSE, UIU
Submission Date: 3 October, 2026

Table of Contents

1.Introduction	4
1.1 Purpose	4
1.2 Intended Audience.....	4
1.3 Scope of the System	5
2.System Overview	8
2.1 Products Perspective	8
2.2 Product Functions	8
2.3 User Classes and Characteristics	12
1. Learner	12
2.4 Operating Environment	14
2.5 Design and Implementation Constraints	15
2.6 Assumptions and Dependencies	16
2.7 SDLC Approach.....	16
3. Requirements Elicitation & Analysis	17
3.1 Benchmark Analysis.....	17
How Superprof works.....	17
3.2 Requirements Analysis	20
3.3 Gap Analysis.....	21
3.4 Feature List Fixation.....	22
4. Functional Requirements	24
5 External & Non-Functional Requirements	25
5.1 User Interfaces (UI).....	25
5.2 Software Interfaces.....	25
5.3 Performance Requirements	25
5.4 Security Requirements	26
5.5 Maintainability & Extension	26
6. Feasibility & Project Management Analysis.....	26
6.1 Feasibility Analysis.....	26
6.2 SWOT Analysis.....	28
7. System Design (Diagrams).....	29
7.1 Data Flow Diagram (DFD)	29
7.2 Use Case Diagram.....	30

8. Prototype Design.....	31
8.1 Screenshots of prototypes- Website	31
8.3 UI Design Principles	44
9. Conclusion.....	45
10. References	45
11. Appendices	46

1.Introduction

1.1 Purpose

SkillSwap is a centralized skill-learning platform that connects learners with suitable mentors based on their skills and location, while providing integrated learning management, communication, progress tracking, payments, reviews, and administrative management. SkillSwap is a skill-learning and mentor–learner matching platform. Its main purpose is to connect people who want to learn a skill with people who can teach that skill, then manage the entire learning relationship through one platform.

1.2 Intended Audience

Our Intended Audiences are:

- **Skill holders/ Mentors:** This system is designed for skill holders so that they can join and teach the learners their skills.
- **Learners :** Learners can efficiently learn any skill they want.
- **Product Sellers:** can use this system to sell products.
- **Project Sponsor & Stakeholders:** To understand the system objectives and verify if requirements meet expectations.
- **System Developers & Designers:** To use as a guideline for system architecture, design, and implementation.
- **Testing Team:** To create test cases and validate if system functionalities are working according to the requirements.
- **System Administrators:** To understand system control, verification, and moderation responsibilities.
- **Future Maintenance Team:** To refer to system requirements for updates and improvements.

1.3 Scope of the System

The scope of the SkillSwap system is quite broad. It is not limited to simply finding mentors; it covers the complete lifecycle of skill-based learning between learners and mentors, together with administration and supporting services.

1. Learner Management

The system covers the learner's activities from registration through completion of a learning relationship.

- Account registration and authentication
- Terms & regulations acceptance
- Search for mentors by skill and location
- View mentor profiles, skills, pricing, experience, and ratings
- Send learning requests
- Communicate with mentors through chat
- View learning plans
- Track learning progress
- Receive invoices and make demo payments
- View payment/invoice history
- Review and rate mentors
- Receive system notifications
- Use premium services and the store

2. Mentor Management

SkillSwap also covers the complete mentor side of the platform.

- Mentor registration and verification
- Mentor profile creation
- Skill and pricing management
- Accept/reject learner requests
- Manage active learners
- Create and manage learning plans
- Update learner progress
- Communicate with learners
- Generate one-time, weekly, or monthly invoices
- Track paid, pending, and overdue invoices
- Calculate earnings
- Request withdrawals
- View ratings and reviews
- Sell products through the integrated store

3. Mentor Discovery & AI Recommendation

A major part of the system is helping learners find appropriate mentors.

The system supports:

- Skill-based mentor search
- Location-based search
- Experience-based matching
- Distance calculation
- Mentor recommendation logic
- AI-assisted mentor recommendation through an API

So the scope includes both manual discovery and automated recommendation.

4. Communication & Notification

The scope includes communication between mentors and learners through:

- One-to-one chat
- Learning-related notifications
- Request notifications
- Payment notifications
- Progress notifications
- Review notifications
- Other account notifications

5. Payment & Financial Management

The system includes an internal financial-management layer:

- One-time payments
- Weekly billing
- Monthly billing
- Invoice generation
- Due dates
- Overdue invoices
- Mentor earnings
- Platform commission
- Withdrawal requests
- Withdrawal history
- Payment records

6. E-Commerce / Store

The scope also extends to an integrated store where users can:

- Browse products
- Add products to cart
- Checkout
- View orders
- Track order history
- Manage product stock

Mentors can also sell products through the platform.

7. Administrative Management

- User management
- Account approval/rejection
- Mentor verification
- Skill management
- Review management
- Monitoring mentor–learner relationships
- Monitoring learning progress
- Payment monitoring
- Revenue monitoring
- Withdrawal management
- Premium-plan management
- Store/product/order management
- Platform activity monitoring

8. Security

Security is also within the system scope:

- Password hashing
- Session-based authentication
- Role-based authorization
- CSRF protection
- Account approval/suspension
- Verification-document protection
- Upload validation
- Sensitive payment information masking

2. System Overview

2.1 Products Perspective

From a product perspective, SkillSwap is a standalone web-based platform that integrates learners, mentors, and administrators in a single ecosystem. It combines mentor discovery, learning management, communication, progress tracking, payments, reviews, e-commerce, AI recommendations, and administrative services to manage the complete skill-learning lifecycle.

2.2 Product Functions

1. User Registration & Authentication

- User registration and account creation
- Login and logout
- Role-based access for **Learners, Mentors, and Administrators**
- Profile management
- Password management
- Account verification and approval
- Account suspension/deactivation

2. Mentor Discovery & Search

Learners can find mentors based on their requirements.

- Search mentors by skill
- Search mentors by location
- Filter mentors based on available information
- View mentor profiles
- View mentor skills and experience
- View mentor pricing
- View mentor ratings and reviews
- Calculate/display relevant distance information
- Receive AI-assisted mentor recommendations

3. Mentor Profile & Skill Management

Mentors can maintain their professional information.

- Create and update mentor profiles
- Add/remove skills
- Set skill-specific pricing

- Provide experience information
- Upload relevant verification documents
- Manage mentor availability/status
- Display ratings and reviews

4. Learning Request Management

The system facilitates the initial connection between a learner and mentor. Functions include:

- Send learning requests
- View sent requests
- Accept requests
- Reject requests
- Cancel requests
- Track request status
- Establish active mentor–learner relationships

5. Learning Management

After a request is accepted, the system supports the learning process.

- Create learning plans
- Define learning goals
- Define milestones/tasks
- Set target dates
- Track completion
- Update learning progress
- View current learning status
- Manage active learners

6. Communication

SkillSwap provides communication between mentors and learners.

- One-to-one messaging
- Conversation management
- Message history
- Contact/access control
- Notifications for relevant activities

7. Progress Tracking

The system allows mentors and learners to monitor learning development.

- Record learning activities

- Update task/milestone status
- Track percentage of completion
- View learning progress
- Monitor completed and pending activities

8. Payment & Invoice Management

The system provides financial-management functionality for the mentor–learner relationship.

- Create invoices
- Support different billing periods
 - One-time
 - Weekly
 - Monthly
- View invoice details
- Track payment status
- Record demo payments
- Calculate mentor earnings
- Calculate platform commission
- Manage withdrawal requests
- Maintain payment history

Note: The current project uses payment functionality for demonstration/internal processing rather than a live banking or payment-gateway system.

9. Review & Rating System

After interacting with a mentor, learners can evaluate their experience.

- Submit ratings
- Write reviews
- View mentor reviews
- Display mentor ratings
- Manage inappropriate reviews through administration

10. Notification Management

The system provides notifications for important activities.

Examples include:

- New learning request
- Request acceptance/rejection
- New message
- Invoice generated
- Payment status

- Learning-plan updates
- Progress updates
- Reviews
- Account-related events

11. AI Mentor Recommendation

The system can recommend suitable mentors to learners. This reduces the effort required for learners to manually search through mentors.

12. Premium Subscription Management

The system supports premium services.

- Display available plans
- Subscribe to premium plans
- Manage subscription status
- Track subscription information
- Apply premium-related functionality

13. E-Commerce / Store Functions

SkillSwap also contains a store component.

Customer functions:

- Browse products
- View product details
- Add products to cart
- Checkout
- View orders
- View order history

Seller/mentor functions:

- Add products
- Update products
- Manage inventory
- View product orders

14. Administrative Functions

Administrators have centralized control over the platform.

- Manage users

- Manage mentors
- Verify mentor accounts
- Manage skills
- Monitor learner–mentor relationships
- Manage reviews
- Monitor payments
- Manage withdrawals
- Manage premium plans
- Manage products
- Manage orders
- Monitor platform activities
- Manage system settings

15. Security Functions

The system also provides basic security mechanisms.

- Password hashing
- Authentication
- Role-based authorization
- Session management
- CSRF protection
- Secure document/file handling
- Upload validation
- Access restrictions for sensitive information

2.3 User Classes and Characteristics

1. Learner

A **Learner** is a registered user who wants to acquire a new skill from a suitable mentor.

Characteristics:

- Basic computer/web browsing knowledge is sufficient.
- Can search and filter available mentors.
- May have limited knowledge of the skill they want to learn.
- Interacts with mentors through the platform.
- Can have one or multiple active learning relationships depending on system rules.
- Can use learning, payment, review, and notification features.

Major activities:

- Register/login

- Create and manage profile
- Search for skills and mentors
- View mentor profiles
- Receive AI-based mentor recommendations
- Send learning requests
- Communicate with mentors
- View learning plans
- Monitor learning progress
- View and pay invoices
- View payment history
- Submit ratings and reviews
- Manage notifications
- Purchase products
- Manage premium subscription

2. Mentor

A **Mentor** is a user who possesses knowledge or expertise in one or more skills and provides learning services to learners.

Characteristics:

- Has knowledge/experience in specific skills.
- Can create a professional mentor profile.
- Can specify skills and pricing.
- Requires verification/approval where applicable.
- Communicates directly with learners.
- Manages learning plans and learner progress.
- Can manage invoices and earnings.

Major activities:

- Register/login
- Create and manage mentor profile
- Add and manage skills
- Set pricing
- Submit verification information
- Receive learning requests
- Accept/reject learning requests
- Manage active learners
- Communicate with learners
- Create learning plans
- Update learner progress
- Generate/manage invoices
- Monitor earnings
- Request withdrawals

- View reviews and ratings
- Manage products in the store
- Receive notifications

3. Administrator

An **Administrator** is responsible for managing and monitoring the overall SkillSwap platform.

Characteristics:

- Has advanced knowledge of the system.
- Requires a higher level of authorization.
- Responsible for maintaining platform integrity.
- Can access administrative dashboards and management functions.
- Can monitor activities across different user groups.

Major activities:

- Manage users
- Approve/reject or manage accounts
- Verify mentors
- Manage skills
- Monitor mentor–learner relationships
- Manage reviews
- Monitor payments and invoices
- Manage withdrawal requests
- Manage premium plans
- Manage products and orders
- Monitor system activities
- Manage platform-level settings

2.4 Operating Environment

The SkillSwap system operates as a web-based application consisting of a client-side interface, a Flask/Python server, and a MySQL database.

- **Client:** Modern web browsers such as Chrome, Edge, Firefox, or Safari.
- **Frontend:** HTML, CSS, JavaScript, and Bootstrap, jinja2
- **Backend:** Python with the Flask framework.
- **Database:** MySQL for storing users, mentors, skills, learning relationships, payments, reviews, products, and other system data.
- **Operating Systems:** Windows.

- **Network:** Internet/local network connection between the client and server.
- **Server Environment:** A Python-compatible server environment; the project also includes Docker/Docker Compose configuration for containerized deployment.
- **External Services:** The system can communicate with supporting services such as the AI recommendation service.

2.5 Design and Implementation Constraints

The SkillSwap system has several constraints that must be considered during design and implementation:

- **Technology Constraint:** The system is implemented using Python/Flask, web technologies, and MySQL, so the design must remain compatible with these technologies.
- **Database Constraint:** User, mentor, learning, payment, review, and store data must be properly structured and maintained in the MySQL database.
- **Security Constraint:** Authentication, authorization, password hashing, session management, CSRF protection, and secure file handling must be maintained.
- **Role-Based Access:** Different functions must be restricted according to Learner, Mentor, and Administrator roles.
- **Internet Dependency:** The web application requires network connectivity for client-server communication and external services.
- **Payment Constraint:** The current system provides demo/internal payment functionality rather than direct real-world banking or payment-gateway transactions.
- **AI Dependency:** AI-based mentor recommendations depend on the availability and proper configuration of the external AI service/API.
- **Browser Compatibility:** The interface should work correctly on commonly used modern web browsers.
- **Responsive Design:** The interface should remain usable across desktop, tablet, and mobile screen sizes.
- **Data Privacy:** Personal information, mentor verification documents, payment information, and other sensitive data must be protected from unauthorized access.
- **Deployment Constraint:** The system must operate within the supported Python environment and can use the provided Docker/Docker Compose configuration for deployment.

2.6 Assumptions and Dependencies

Assumptions

The SkillSwap system is developed based on the following assumptions:

- Users have a stable internet connection and access to a modern web browser.
- Learners provide accurate information about the skills they want to learn.
- Mentors provide accurate information about their skills, experience, pricing, and availability.
- Users provide valid registration and profile information.
- Administrators are responsible for verifying mentors and managing inappropriate or incorrect information.
- Users have basic knowledge of using web-based applications.
- The MySQL database remains available and properly configured.
- Users are responsible for maintaining the confidentiality of their login credentials.
- Payment information used in the current system is assumed to be demonstration/internal data, rather than real financial transactions.

Dependencies

The system depends on:

- Python and Flask for backend operation.
- MySQL for database storage and management.
- Modern web browsers for accessing the application.
- Internet/network connectivity for communication between clients and the server.
- External AI service/API for AI-based mentor recommendations.
- Required Python packages and libraries specified by the project.
- Docker/Docker Compose, where containerized deployment is used.
- File-storage mechanisms for profile images and mentor verification documents

2.7 SDLC Approach

The SkillSwap system follows an Agile SDLC approach. Agile is suitable because the system is developed in small, incremental iterations, allowing requirements, features, and UI designs to be refined throughout development.

Main Agile Phases

1. **Requirement Analysis** – Identify requirements of learners, mentors, and administrators.
2. **Planning** – Divide the system into modules such as authentication, mentor management, learning management, payments, and administration.
3. **Design** – Design the database, system architecture, UI, DFD, use-case diagrams, and module structure.
4. **Implementation** – Develop the system module by module using Flask/Python, MySQL, HTML, CSS, JavaScript, etc.
5. **Testing** – Perform functional, integration, and system testing for each implemented module.
6. **Review & Feedback** – Evaluate the implemented features and identify required improvements.
7. **Iteration** – Modify and improve the system based on testing and feedback.
8. **Deployment & Maintenance** – Deploy the completed system and fix bugs or enhance features when required.

3. Requirements Elicitation & Analysis

3.1 Benchmark Analysis

1. Superprof

Superprof is an online tutoring marketplace that connects students with tutors, teachers, coaches, and other skilled people. Users can find tutors for academic subjects, languages, programming, music, sports, arts, hobbies, career development, and many other areas.

How Superprof works

1. Search for a tutor

A student selects what they want to learn and can search by subject and location, including online lessons.

2. Compare tutor profiles

Students can look at tutor information such as:

- Skills/subjects
- Experience and qualifications
- Hourly price
- Reviews and ratings
- Location
- Online or in-person availability

3. Contact the tutor

The student sends a lesson request or contacts the selected tutor. The student generally chooses the tutor rather than having one automatically assigned.

4. Arrange the lessons

The student and tutor discuss their requirements, availability, lesson schedule, and learning method. Lessons can be **online or face-to-face**.

5. Learn and review

The student takes lessons and can provide feedback/reviews about the tutor. Tutors can also manage their own teaching conditions, including their rates and schedules.

2. MentorCruise

MentorCruise is an online 1-on-1 mentorship platform that connects people who want to learn or improve their careers with experienced professionals and industry mentors.

How it works

The basic process is:

1. Find a mentor

Users search for mentors based on company, skills, role, or career area such as software engineering, UX design, data science, product management, marketing, and leadership.

2. View mentor profiles

A mentor profile can show their professional background, skills, ratings, and mentorship offerings. The platform describes its mentors as vetted professionals.

3. Choose a mentorship plan

Users can choose flexible mentorship arrangements, including personal chats, regular calls, or other structures. The platform also offers monthly mentorship and individual sessions.

4. Learn and communicate

The learner works directly with the mentor through 1-on-1 communication, calls, feedback, and personalized guidance.

5. Continue long-term mentorship

The platform focuses particularly on ongoing career mentorship, but it also offers one-off sessions such as introductory calls, study plans, and interview preparation.

3. Preply

Preply is an online tutoring platform, mainly focused on language learning. It connects learners with tutors from around the world for personalized 1-to-1 online lessons.

How Preply works

1. Find a tutor

Learners search for tutors based on factors such as:

- Language/subject
- Price
- Availability
- Specialization
- Qualifications
- Native language/accent
- Reviews and ratings

2. View tutor profiles

Learners can check a tutor's profile, introduction video, experience, teaching style, ratings, and verified reviews before choosing them.

3. Take a trial lesson

Learners can book a 25- or 50-minute trial lesson with a tutor through Preply's online learning environment.

4. Continue with regular lessons

If the learner likes the tutor, they can subscribe to a flexible lesson plan and schedule recurring lessons according to their needs.

5. Track learning and practice

Preply provides an online classroom and additional learning tools, including AI-assisted practice, lesson insights, vocabulary practice, and personalized exercises.

comparison table

Feature	 SkillSwap	MentorHub	 Udemy	Superprof	 MentorCruise	Preply 
Free Teaching Option	✓	✓	✓	✓	✓	✓
Paid Teaching Option	✓	✓	✓	✓	✓	✓
Nearby Mentor Search	✓	✗	✗	Limited	✗	✗
Skill Exchange Between User	✓	✗	✗	✗	✗	✗
Document Verification	✓	✓	✓	✓	✓	✓
In-App-Chat	✓	✓	✗	✓	✓	✓
Rating & Review	✓	✓	✓	✓	✓	✓
Learning Equipment Store	✓	✗	✗	✗	✗	✗
Community Based Learning	✓	✗	✗	✗	✓	✗

3.2 Requirements Analysis

Requirements gathering is the foundational blueprinting phase of the system development life cycle (SDLC). It bridges the gap between a high-level business concept and a working software platform.

Purpose

- **Eliminating Ambiguity:** To translate raw ideas into clear, concrete functional specifications.
- **Risk Reduction:** Catching conflicting logic early—such as resolving how the consolidated **Organizer** / **Subscriber** / **Seller** role balances hosting events with inventory tracking—before writing code.
- **Cost & Time Efficiency:** Preventing expensive structural updates or codebase overhauls down the line by cementing data flows and relational models first.
- **Stakeholder Alignment:** Ensuring software developers, project managers, system administrators, and target gaming communities are all working toward the exact same outcome.

3.3 Gap Analysis

Area	Existing Platforms	Gap Identified	SkillSwap Approach
Mentor/Tutor Discovery	Most platforms provide search and filtering	Finding a mentor based on multiple contextual factors can be fragmented	Skill and location-based mentor discovery
Nearby Mentor Search	Limited or not central to many platforms	Users may not easily find mentors available near them	Location-based mentor search
Skill Exchange	Most platforms focus on paid tutoring/mentoring	Little emphasis on users exchanging skills with each other	Free skill exchange option
Learning Relationship Management	Often focused on booking individual lessons	Long-term mentor–learner relationships may require external tools	Learning relationship management
Learning Plans	Available to varying degrees	Learning activities may not be managed in one integrated workflow	Mentor-created learning plans
Progress Tracking	Not consistently integrated across platforms	Learners may need separate tools to track progress	Built-in progress tracking
Communication	Some platforms provide messaging; others rely on external communication	Communication can be fragmented	In-app mentor–learner chat
AI Mentor Recommendation	Some platforms have matching/recommendation mechanisms	Recommendation approaches and available factors vary	AI-assisted mentor recommendation
Mentor Verification	Some platforms use verification	Verification methods differ between platforms	Document-based mentor verification
Payments & Learning Management	Payment is commonly available in tutoring marketplaces	Learning, invoices, and mentor earnings may be separated	Integrated invoices, payments, earnings, and withdrawals
Learning Equipment Store	Usually not part of mentoring platforms	Learners may need another platform to obtain learning equipment	Integrated store
Community-Based Learning	Many platforms emphasize one-to-one tutoring	Skill exchange/community learning may receive less emphasis	Community-oriented skill exchange
Administration	Platforms have internal administration	The scope of administrative controls differs	Centralized admin dashboard

3.4 Feature List Fixation

USER Interface

2. Learner Features

- Learner profile
- Search for mentors
- Search by skill
- Search by location
- View mentor profiles
- View mentor skills and experience
- View mentor pricing
- View ratings and reviews
- AI-based mentor recommendations
- Send learning requests
- Track request status
- Manage active mentors
- Chat with mentors
- View learning plans
- Track learning progress
- View invoices
- Make demo payments
- View payment history
- Rate and review mentors
- Receive notifications
- Premium subscription
- Browse learning products
- Shopping cart and checkout
- Order history

3. Mentor Features

- Mentor registration
- Mentor profile creation
- Skill management
- Pricing management
- Experience/profile information
- Document verification
- Learner request management
- Accept/reject learning requests

- Active learner management
- One-to-one chat
- Create learning plans
- Manage learning milestones
- Update learner progress
- Invoice generation
- One-time/weekly/monthly billing
- Earnings tracking
- Platform commission calculation
- Withdrawal requests
- Withdrawal history
- View ratings and reviews
- Notifications
- Premium services
- Product/store management

Admin Interface

13. Admin Features

- Admin dashboard
- User management
- Learner management
- Mentor management
- Mentor verification
- Skill management
- Learning relationship monitoring
- Review management
- Payment monitoring
- Invoice monitoring
- Revenue monitoring
- Withdrawal management
- Premium-plan management
- Product management
- Order management
- Platform activity monitoring

4. Functional Requirements

These describe **what the system should do**.

User & Authentication

- The system shall allow users to register and log in.
- The system shall support different roles: **Learner, Mentor, and Administrator**.
- The system shall manage user profiles and account status.
- The system shall require acceptance of Terms & Regulations during registration.

Mentor Discovery

- Learners shall be able to search mentors by **skill and location**.
- Learners shall be able to view mentor profiles, skills, pricing, ratings, and relevant information.
- The system shall provide **AI-assisted mentor recommendations**.

Learning Management

- Learners shall be able to send learning requests to mentors.
- Mentors shall be able to accept or reject requests.
- The system shall create an active mentor–learner relationship after acceptance.
- Mentors shall be able to create learning plans.
- Learners and mentors shall be able to view learning progress.

Communication

- Learners and mentors shall be able to communicate through **in-app chat**.
- The system shall provide notifications for important activities.

Payment & Financial Functions

- Mentors shall be able to issue one-time, weekly, or monthly invoices.
- Learners shall be able to view and pay demo invoices.
- The system shall track paid, pending, and overdue invoices.
- The system shall calculate mentor earnings and platform commission.
- Mentors shall be able to submit withdrawal requests.

Reviews

- Learners shall be able to rate mentors using a **1–5 star rating system**.
- Learners shall be able to submit reviews.
- Users shall be able to view mentor ratings and reviews.

Store

- Users shall be able to browse products.
- Users shall be able to add products to a cart and place orders.
- Mentors shall be able to sell products.
- The system shall manage product stock and order history.

Administration

- Administrators shall be able to manage users and accounts.
- Administrators shall be able to verify mentors.
- Administrators shall be able to manage skills and reviews.
- Administrators shall be able to monitor mentor–learner relationships.
- Administrators shall be able to monitor payments, revenue, withdrawals, premium plans, products, and orders.

5 External & Non-Functional Requirements

5.1 User Interfaces (UI)

- **Tailored Layouts:** Custom dashboards optimized by role
- **Fast Navigation:** Sidebar controls allowing users to switch between main modules in 2 clicks or fewer.
- **Fluid Responsiveness:** Adaptive styling (Tailwind/Bootstrap) to ensure smooth layouts across mobile, tablet, and desktop viewports.

5.2 Software Interfaces

- **Payment Processing APIs:** Secure connections to regional mobile wallets (**bKash, Nagad, Rocket**) and card aggregators using asynchronous webhooks.
- **Notification Bridges:** Standard SMTP and external SMS gateways to dispatch automated verification OTPs and bracket alerts.
- **RDBMS Connectors:** Secure Object-Relational Mapping (ORM) drivers to handle safe database communications without locking tables.

5.3 Performance Requirements

- **Low Latency:** Public directories must load under **2.0 seconds**; high-frequency events (like in-app messaging) must display within **500 milliseconds**.
- **High Scalability:** Built to handle up to **10,000 concurrent connections** during peak tournament timelines.
- **High Availability:** A minimum uptime baseline of **99.9%** to prevent database race conditions during registration deadlines.

5.4 Security Requirements

- **Role-Based Restrictions:** Strict access rules applied at both the front-end and API routing levels to prevent unauthorized actions.
- **Session Safeguards:** Secure password encryption via bcrypt coupled with cryptographically signed session tokens (JWT) to stop hijacking attempts.
- **Data Privacy:** Full TLS 1.3 (HTTPS) encryption applied to all network communications safeguarding personal and financial records.

5.5 Maintainability & Extension

- **Normalized Data Model:** Clean structural separation of tables to support seamless database upgrades without data redundancy.
- **Modular Codebase:** Disconnected functional components (e.g., Billing separate from the Bracket Engine) to safely isolate code changes.
- **Immutable Audit Ledger:** A dedicated AdminLog table that logs all supervisor activities, simplifying system tracking and bug debugging.

6. Feasibility & Project Management Analysis

6.1 Feasibility Analysis

1. Technical Feasibility

- Uses commonly available technologies such as Python/Flask, MySQL, HTML, CSS, and JavaScript.
- The development team can implement the required web-based features using these technologies.
- AI recommendation functionality can be integrated through an external AI service/API.
- Docker/Docker Compose can support deployment.
- No specialized hardware is required.

2. Economic/Financial Feasibility

- The main development technologies are open-source or freely available.
- Development can be performed using existing computers and software tools.
- A basic deployment does not require expensive infrastructure.
- The system can potentially generate revenue through premium subscriptions, platform commissions, and other services.

3. Operational Feasibility

- The system provides separate functionality for learners, mentors, and administrators.
- Users can access the platform through a web browser.
- The interface is designed around familiar activities such as searching, messaging, learning management, payments, and reviews.
- Centralized administration makes it possible to manage users, mentors, skills, payments, and platform activities.

4. Human Resources Feasibility

- **Development Phase:** Requires a small, standard team of web developers (Frontend, Backend, and Database Engineers) familiar with relational architectures and API integrations. No highly specialized technical positions (like AI or hardware scientists) are required.
- **Post-Launch Phase:** Operations require a lean administrative team to monitor the platform, handle customer support tickets via the messaging hub, and review vendor/hosting verification requests.

5. Market / Legal / Political Feasibility

The system is technically capable of implementing basic security measures such as:

- Password hashing
- Role-based access control
- Authentication
- CSRF protection
- Secure file handling
- Protection of sensitive user information

However, real-world deployment would require additional privacy, payment, consumer-protection, and data-protection compliance depending on the target country and services used.

6. Schedule Feasibility

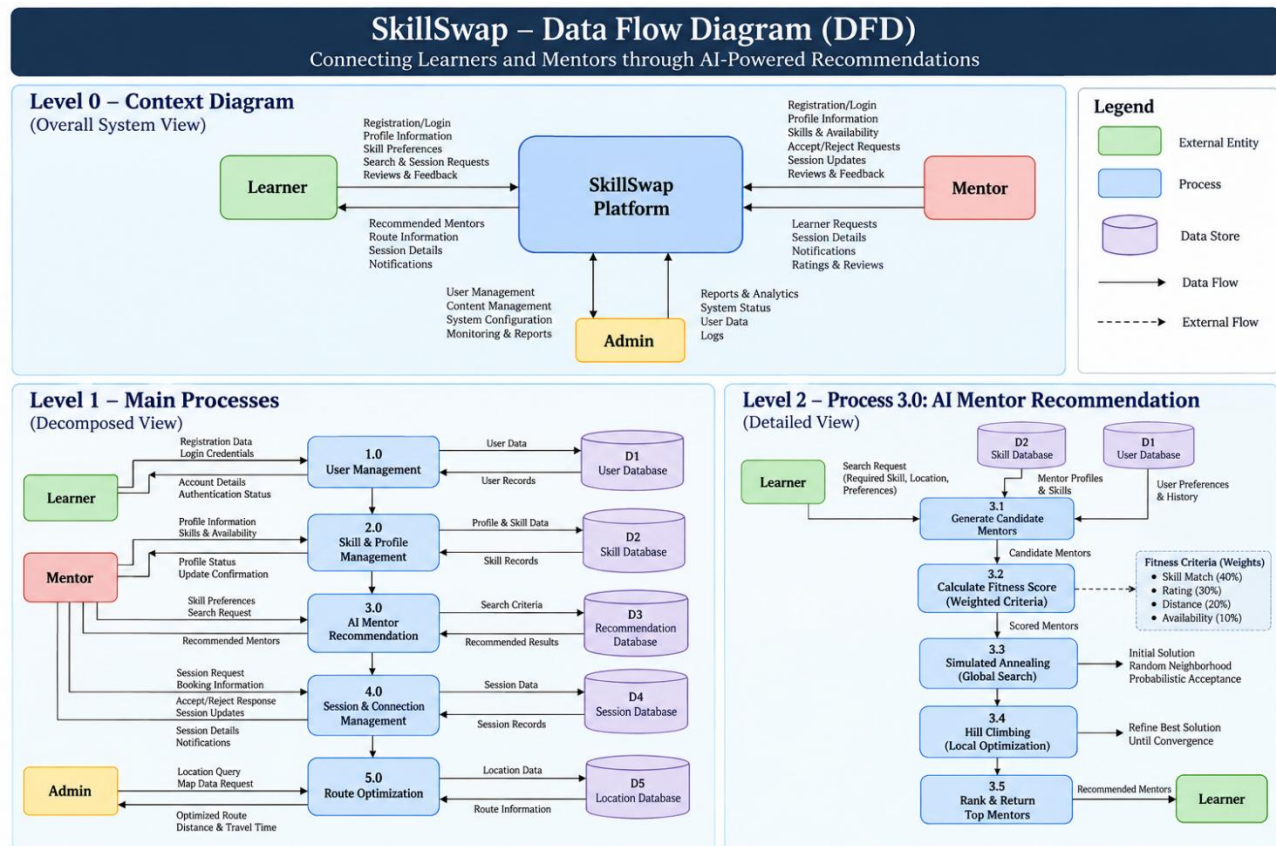
- **Modular Development Track:** Because the system features distinct, independent modules (e.g., Billing, Bracket Engines, Storefront) that align perfectly with a Role-Based Access Control model, development tasks can run simultaneously.
- **Timeline Efficiency:** Teams can code the front-end user interfaces, backend API routes, and relational schemas in parallel, allowing a fully operational MVP (Minimum Viable Product) to launch comfortably within standard software deployment schedules (typically 3 to 6 months).

6.2 SWOT Analysis

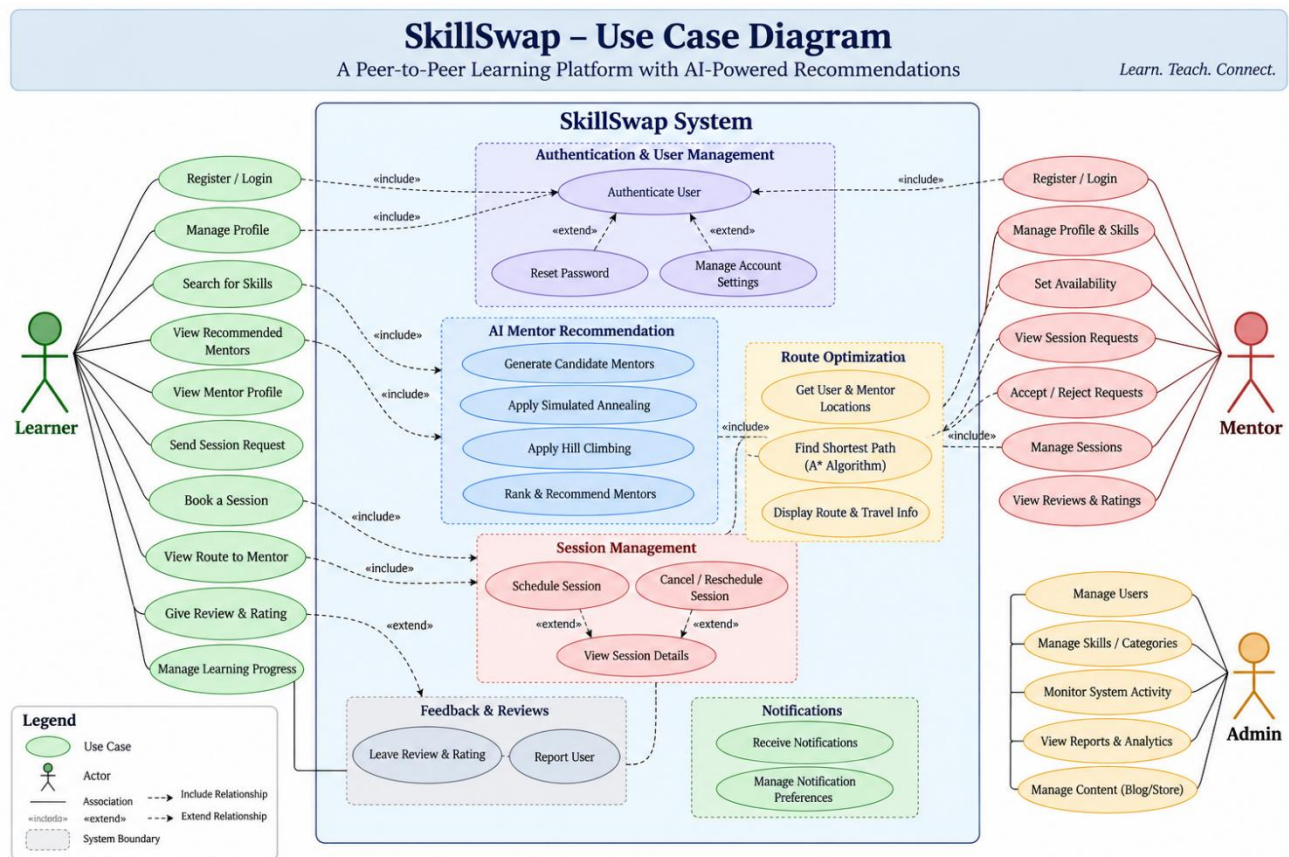
Strengths	Weakness
• Centralized system for Sharing skill with learners • Easy to use for beginners and mentors • Automates learning progress , mentor searching , ai based mentor recommendation. • Reduces manual errors and saves time • Designed specifically for Skill sharing.	• Limited advanced features compared to professional platforms • Basic UI/UX design • Limited scalability for user tracking • Needs better security and performance optimization • realtime user testing and feedbacks
Opportunities	Threats
• Create chances to earning by skill sharing • Can be expanded into a commercial or SaaS platform. • Integration with learning progress tracking . • Collaboration with skill sharing communities and organizations • Future mobile app development possibility	• Competition with platforms like superprof and MentorHub • Increasing user expectations for modern features • Risk of data security issues • Dependence on server and internet stability • Rapid technological changes

7. System Design (Diagrams)

7.1 Data Flow Diagram (DFD)

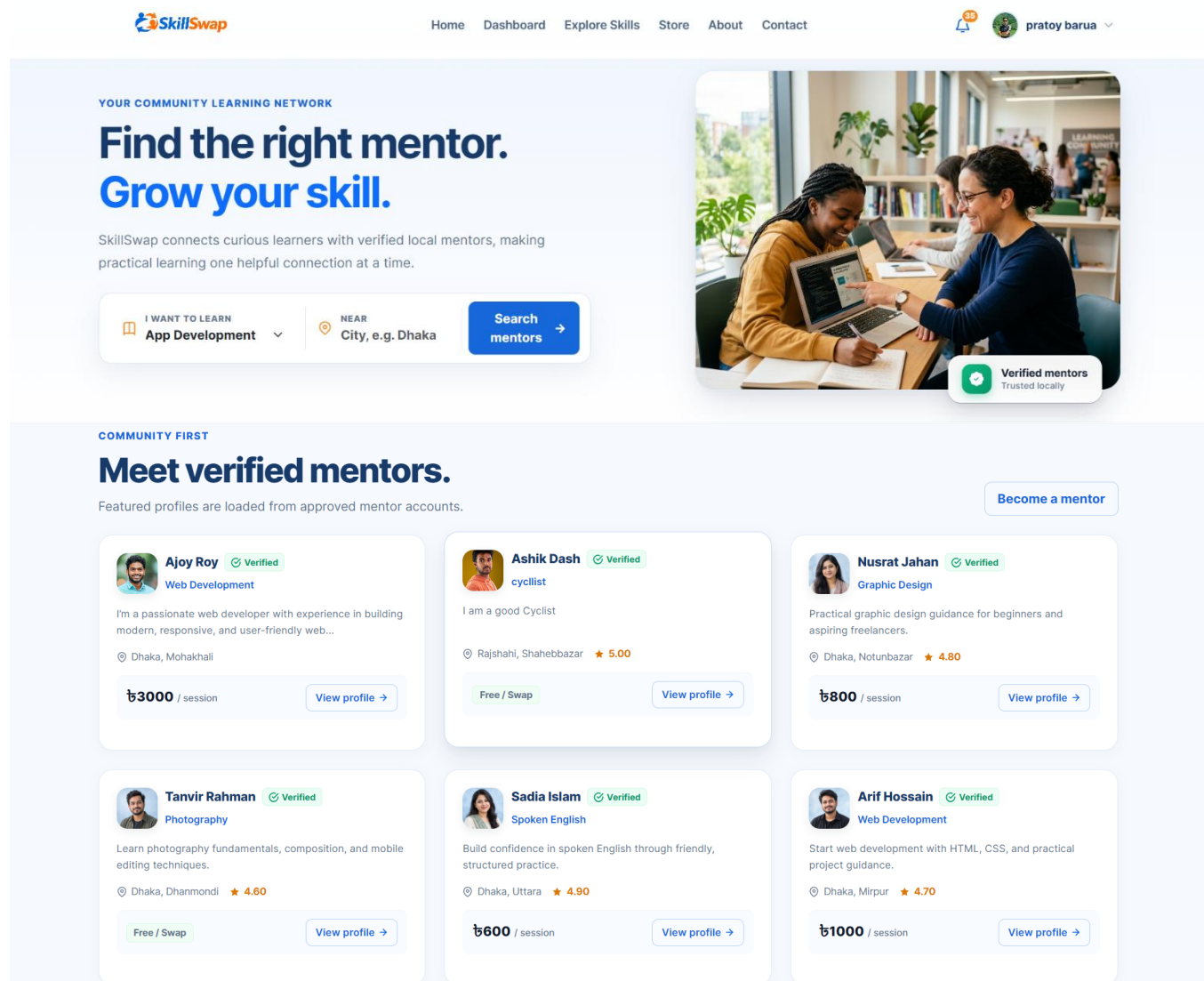


7.2 Use Case Diagram



8. Prototype Design

8.1 Screenshots of prototypes- Website Landing Page



Sign up



Home Explore Skills Store About Contact

[Log in](#) [Sign up](#)

JOIN THE COMMUNITY

How would you like to join?



I want to learn

Find practical skills and trusted mentors nearby.

[Register as learner](#)



I want to mentor

Share your experience with learners in your community.

[Register as mentor](#)

 Explore Join Support Follow us

Skills Become a learner Contact Facebook

Sign in



Home Explore Skills Store About Contact

[Log in](#) [Sign up](#)

WELCOME BACK

Log in to SkillSwap

Learners and mentors can access their account after admin approval.

Email

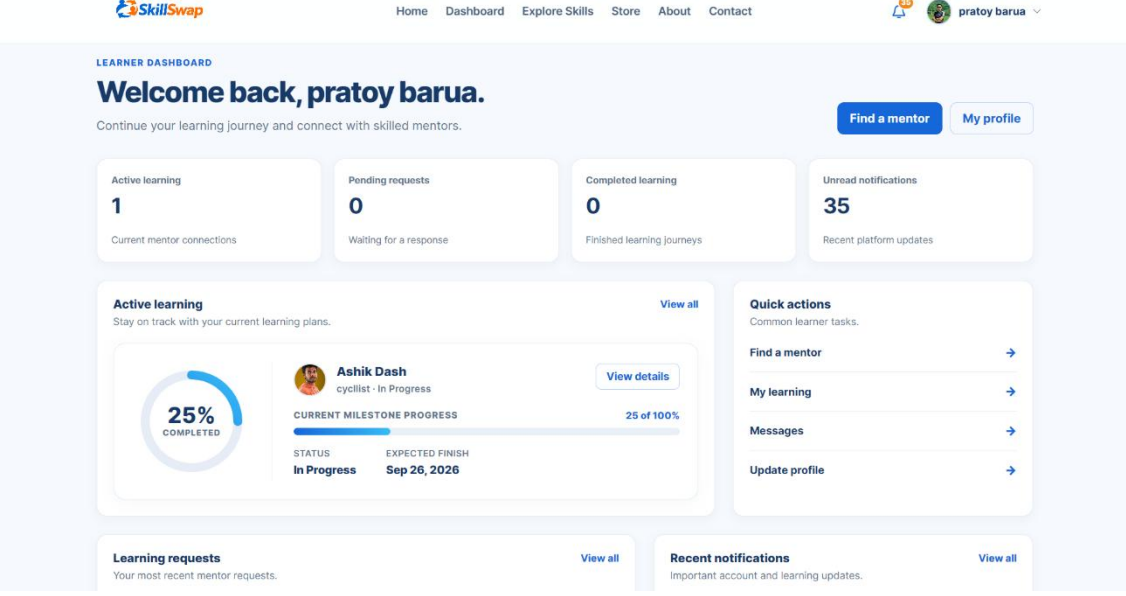
Password

[Log in](#)

Need an account? [Choose registration](#)

027.0.0.145000

User Dashboard



The User Dashboard for SkillSwap features a top navigation bar with links to Home, Dashboard, Explore Skills, Store, About, and Contact. The user's name, prato barua, is displayed in the top right corner. The dashboard is titled "LEARNER DASHBOARD" and welcomes the user back. It includes a "Find a mentor" button and a "My profile" link. The main content area is divided into several sections: a summary of learning progress (Active learning: 1, Pending requests: 0, Completed learning: 0, Unread notifications: 35), an "Active learning" section showing a progress bar for "Ashik Dash" (cyclist) at 25% completion, and a "Quick actions" section with links to Find a mentor, My learning, Messages, and Update profile. At the bottom, there are sections for "Learning requests" and "Recent notifications".

LEARNER DASHBOARD

Welcome back, prato barua.

Continue your learning journey and connect with skilled mentors.

[Find a mentor](#) [My profile](#)

Active learning
1
Current mentor connections

Pending requests
0
Waiting for a response

Completed learning
0
Finished learning journeys

Unread notifications
35
Recent platform updates

Active learning
Stay on track with your current learning plans. [View all](#)

Ashik Dash
cyclist · In Progress [View details](#)

25% COMPLETED

CURRENT MILESTONE PROGRESS 25 of 100%

STATUS In Progress **EXPECTED FINISH** Sep 26, 2026

Quick actions
Common learner tasks.

[Find a mentor](#) →

[My learning](#) →

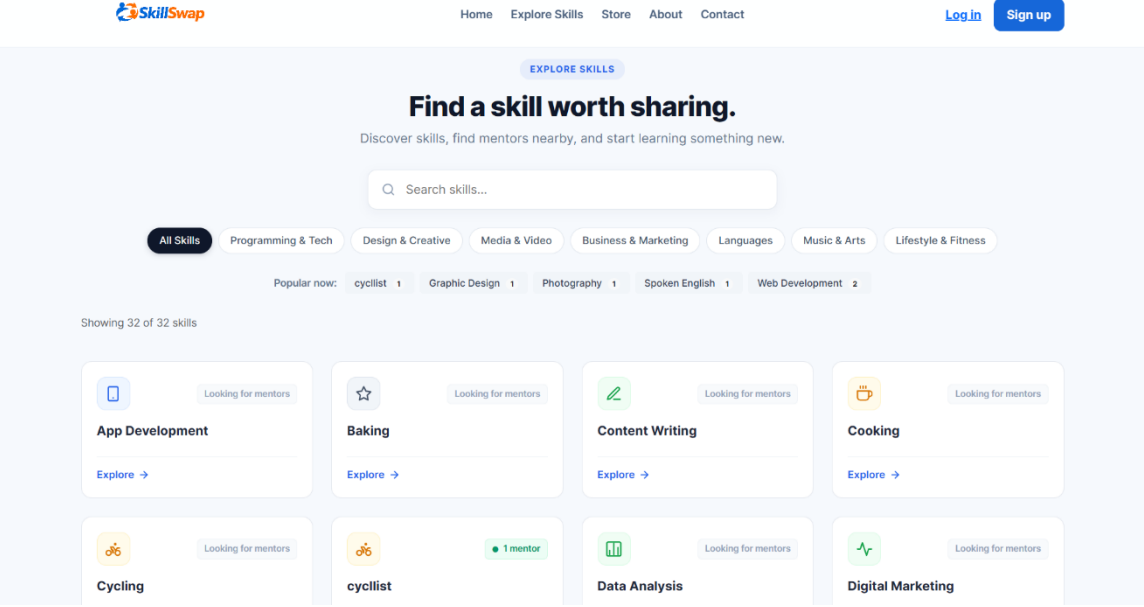
[Messages](#) →

[Update profile](#) →

Learning requests
Your most recent mentor requests. [View all](#)

Recent notifications
Important account and learning updates. [View all](#)

Learner Find SKill



The Learner Find Skill page features a top navigation bar with links to Home, Explore Skills, Store, About, and Contact. The user is logged in, as indicated by the "Log In" and "Sign up" buttons. The page is titled "EXPLORE SKILLS" and encourages users to "Find a skill worth sharing." It includes a search bar and a list of skill categories: All Skills, Programming & Tech, Design & Creative, Media & Video, Business & Marketing, Languages, Music & Arts, and Lifestyle & Fitness. Below the categories, a "Popular now" section lists skills like cyclist, Graphic Design, Photography, Spoken English, and Web Development. The main content area displays a grid of skill cards, each with an icon, a title, and a "Looking for mentors" status. The skills shown are App Development, Baking, Content Writing, Cooking, Cycling, cyclist, Data Analysis, and Digital Marketing.

EXPLORE SKILLS

Find a skill worth sharing.

Discover skills, find mentors nearby, and start learning something new.

Search skills...

All Skills Programming & Tech Design & Creative Media & Video Business & Marketing Languages Music & Arts Lifestyle & Fitness

Popular now: cyclist 1 Graphic Design 1 Photography 1 Spoken English 1 Web Development 2

Showing 32 of 32 skills

App Development
Looking for mentors
[Explore →](#)

Baking
Looking for mentors
[Explore →](#)

Content Writing
Looking for mentors
[Explore →](#)

Cooking
Looking for mentors
[Explore →](#)

Cycling
Looking for mentors

cyclist
1 mentor

Data Analysis
Looking for mentors

Digital Marketing
Looking for mentors

AI mentor Recommendation

The screenshot shows the 'AI-POWERED EXPLORATION' section of the SkillSwap website. It features a map of Dhaka, Bangladesh, with a blue dot indicating the user's current location at 23.7944° N, 90.4358° E. To the right of the map, there are two mentor profiles:

- Arif Hossain** (AI Best Match, 76% Match): Web Development - Mirpur, Dhaka. Start web development with HTML, CSS, and practical project guidance.... Distance: 7.81 km, Rating: 4.7 / 5, Pricing: \$1000, Experience: 3 years. Overall Match Competibility: 76%.
- Ajoy Roy** (Nearest Mentor, 5.03 km away): Web Development - Mohakhali, Dhaka. I'm a passionate web developer with experience in building modern, responsive, and user-friendly...

At the top, there are navigation links: Home, Dashboard, Explore Skills, Store, About, Contact. The user's name 'pratoy barua' is visible in the top right corner.

Learner Profile

The screenshot shows the 'YOUR PROFILE' section of the SkillSwap website. It features a profile card with a circular profile picture of a man wearing sunglasses. Below the profile picture, there are two main sections: 'PERSONAL INFORMATION' and 'LOCATION'.

PERSONAL INFORMATION

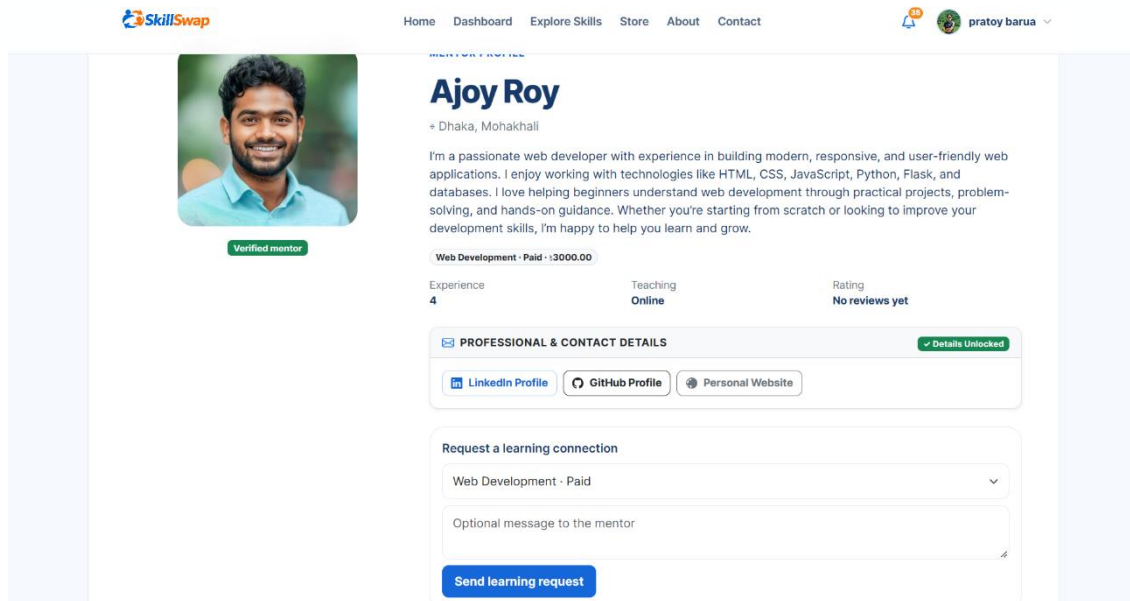
- Full name *: pratoy barua
- Phone *: 01741215105
- Date of birth: mm/dd/yyyy
- Gender: — select —

LOCATION

- City *:
- Area:

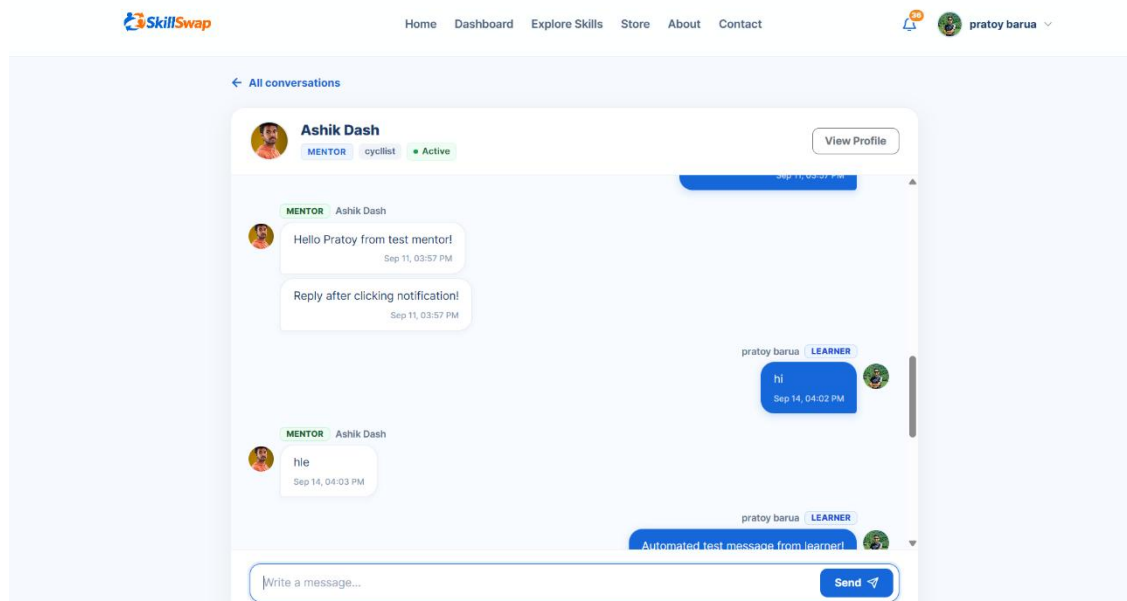
At the top, there are navigation links: Home, Dashboard, Explore Skills, Store, About, Contact. The user's name 'pratoy barua' is visible in the top right corner.

Mentor Profile



The screenshot shows a user profile for Ajoy Roy on the SkillSwap platform. The profile includes a profile picture, a 'Verified mentor' badge, and a bio. The bio states: 'I'm a passionate web developer with experience in building modern, responsive, and user-friendly web applications. I enjoy working with technologies like HTML, CSS, JavaScript, Python, Flask, and databases. I love helping beginners understand web development through practical projects, problem-solving, and hands-on guidance. Whether you're starting from scratch or looking to improve your development skills, I'm happy to help you learn and grow.' The profile also shows 'Web Development - Paid - 13000.00', 'Experience 4', 'Teaching Online', and 'Rating No reviews yet'. There is a section for 'PROFESSIONAL & CONTACT DETAILS' with links to LinkedIn Profile, GitHub Profile, and Personal Website. Below this is a 'Request a learning connection' form with a dropdown menu set to 'Web Development - Paid', an optional message field, and a 'Send learning request' button.

Messages option



The screenshot shows a messaging interface on the SkillSwap platform. The header includes the SkillSwap logo, navigation links (Home, Dashboard, Explore Skills, Store, About, Contact), and a user profile (pratoy barua). The main content area is titled 'All conversations' and displays a list of messages from a mentor named Ashik Dash. The messages are as follows:

- Mentor: Ashik Dash (Active)
 - Mentor: Hello Pratoy from test mentor! (Sep 11, 03:57 PM)
 - Mentor: Reply after clicking notification! (Sep 11, 03:57 PM)
- Learner: hi (Sep 14, 04:02 PM)
- Mentor: hie (Sep 14, 04:03 PM)
- Learner: Automated test message from learner

The interface includes a 'Write a message...' input field and a 'Send' button.

Store



[Home](#) [Explore Skills](#) [Store](#) [About](#) [Contact](#)

[Log In](#) [Sign up](#)

SKILLSWAP STORE

Tools for your next skill.

Equip yourself with useful tools for learning, creating and practicing.

All categories

▼

Min price

▼

Any stock

▼

Search



CREATIVE TOOLS

Matte Blue Premium Acoustic Guitar -2024 Edition

~With 1x Guitar, 5 Picks, Guitar Belt and Carrying Case ~Top: AAA European...

₺3200.00

19 in stock

View product



LEARNING SUPPLIES

A5 Study Notebook

A compact notebook for course notes, learning plans, and daily practice.

₺180.00

21 in stock

View product



CREATIVE TOOLS

Sketching Pencil Set

A practical pencil set for drawing, design exercises, and creative practice.

₺350.00

13 in stock

View product



TECH ACCESSORIES

Laptop Stand

An adjustable desk stand for comfortable online learning sessions.

₺1200.00

11 in stock

View product

Product Overview



[Home](#) [Explore Skills](#) [Store](#) [About](#) [Contact](#)

[Log In](#) [Sign up](#)

[← Back to Store](#)

Home / Store / Creative tools / Matte Blue Premium Acoustic...



CREATIVE TOOLS

Matte Blue Premium Acoustic Guitar -2024 Edition

₺3200.00

In stock (19 available)

~With 1x Guitar, 5 Picks, Guitar Belt and Carrying Case ~Top: AAA European spruce ~Back & Sides: Indian rosewood AAA (Style 32) ~Neck: Mahogany ~Cutaway: Soft (round) ~Soundhole Rosette: Wooden core (matching back & sides) with purflings ~Headstock: Flat headstock ~Fingerboard: Ebony ~Neck width at nut: 44mm (1.73 inch) ~Scale length: 650mm (25.6 inch) ~Pickup system: L.R. Bags Anthem

Please log in to add items to your cart and checkout.

Log in to purchase

✓ Verified Tool

✓ Community Inventory

✓ Secure Checkout

Product Purchase

The screenshot displays the 'Order is being processed' status for a purchase on SkillSwap. The order is #15, placed on October 1, 2026, at 01:47 PM. The payment was successful, and the order is currently in the 'Processing' stage. The invoice details include the store name 'COMMUNITY LEARNING & STORE', the customer's name 'pratoy barua', and the delivery destination 'Savednagar b block, Vandarir mor, Manarash shunna model'.

Order is being processed
Your order has been received and is waiting for confirmation.

Order Progress Timeline

- Order Placed (PLACED)
- Processing (IN REVIEW)
- Confirmed (PENDING)
- Shipped (UPCOMING)
- Delivered (UPCOMING)

INVOICE
Invoice No: INV-000015
Order ID: #15
Order Date: 01 October 2026, 01:47 PM
Payment Ref: DEMO-FFC34C4644F64338BC459DB50A8B449F
ORDER: PROCESSING | PAYMENT: SUCCESSFUL

CUSTOMER INFORMATION (BILLED TO)
pratoy barua
Email: pratoy16@gmail.com

DELIVERY DESTINATION
pratoy barua
Savednagar b block, Vandarir mor, Manarash shunna model

SkillSwap Progress

The screenshot shows the 'Learning Progress' page for a mentorship journey between pratoy barua (Learner) and Ashik Dash (Mentor). The learner is currently active, and the progress is 25% completed. The current focus is 'Other' (Active learning module). The target date is September 26, 2026, and the plan cadence is weekly with a cost of \$200.00. The learning plan shows a progress bar at 25% of 100%, with an expected finish date of September 26, 2026. The mentor's profile, Ashik Dash, is also visible, showing a verified status and a 5.0 rating.

Learning Progress
Mentorship journey between pratoy barua (Learner) and Ashik Dash (Mentor)

OVERALL PROGRESS
25%
Status: In Progress

CURRENT FOCUS
Other
Active learning module

TARGET DATE
Sep 26, 2026
Started: Sep 11, 2026

PLAN CADENCE
\$200.00
Weekly

Learning Plan & Overall Completion

25% COMPLETED

CURRENT MILESTONE PROGRESS
25 of 100%

Status: In Progress
Expected Finish: September 26, 2026

Ashik Dash
Verified Mentor
★ 5.0 | cyclist | 3

[Message Mentor](#)

[View Full Profile](#)

Admin dashboard

Home Dashboard Explore Skills Store About Contact  Admin

PLATFORM CONTROL

Admin dashboard

Monitor and manage the SkillSwap platform from one place.

Manage usersManage skills

Platform overview

People and account activity across SkillSwap.

Learners
2
Registered learner accounts

Mentors
6
Registered mentor accounts

Active users
9
Approved accounts

Pending users
0
Awaiting verification

Learning

Connections, skills and learner feedback.

Active skills
32

Active relationships
1

Reviews
1

Suspended users
0

Business

Payments, store activity and subscriptions.

Successful payments
39

Net revenue
₹2805

Store products
4

Store orders
11

Admin user management

Home Dashboard Explore Skills Store About Contact  Admin

PLATFORM CONTROL

User management

Dashboard

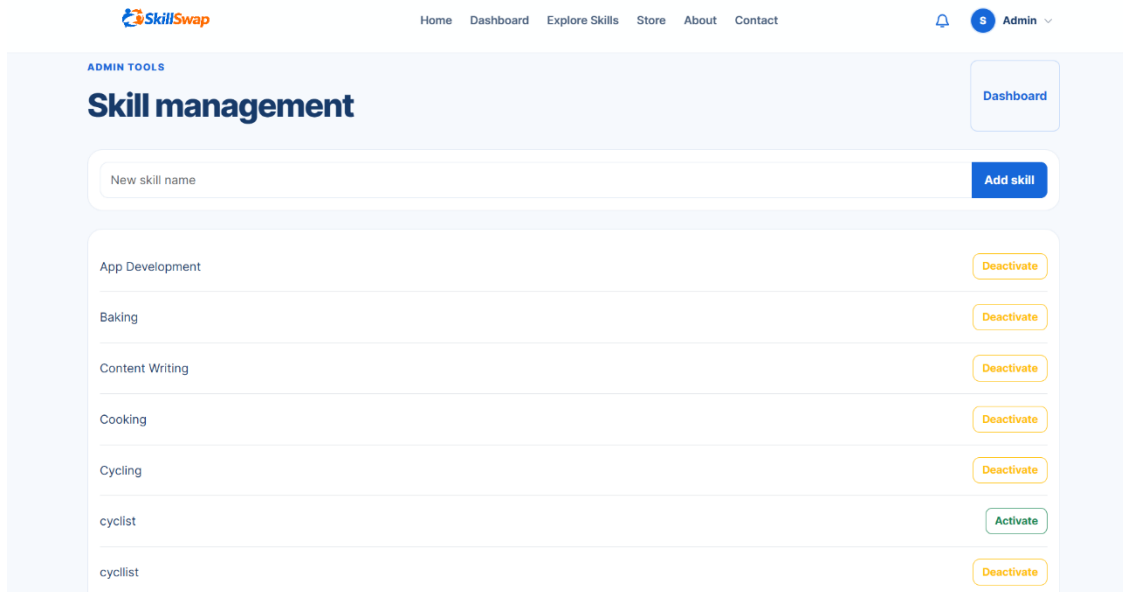
All roles

All statuses

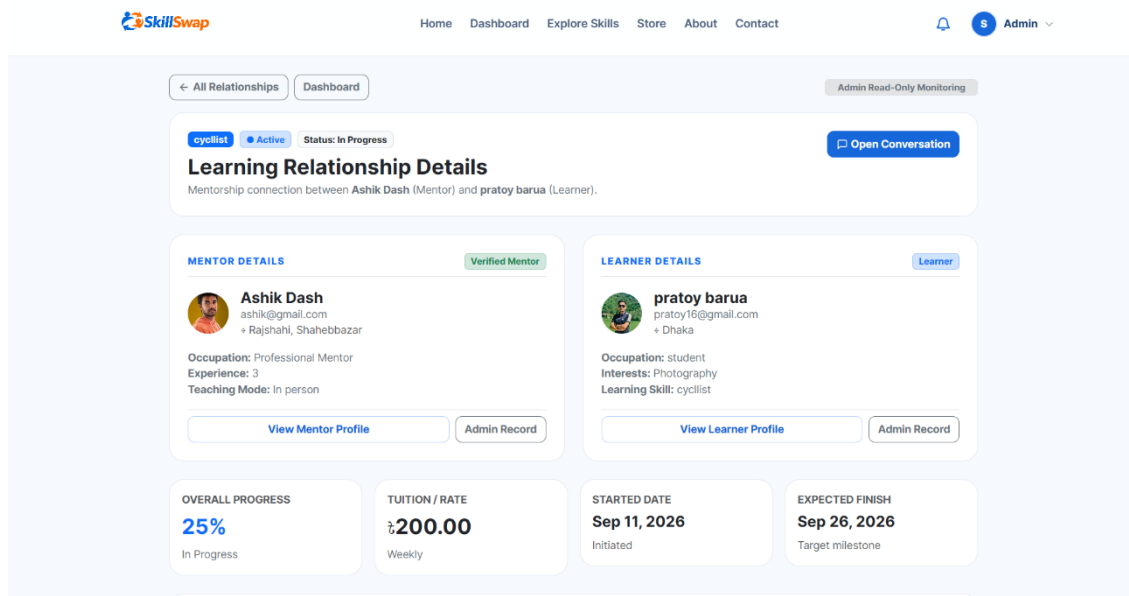
Go

Name	Role	Status	Joined	Actions
Ajoy Roy ajoy16@gmail.com	Mentor	Approved	2026-09-25	Review Delete
Rahim karim rahim@skillswap.com	Learner	Approved	2026-09-14	Review Delete
Ashik Dash ashik@gmail.com	Mentor	Approved	2026-09-11	Review Delete
Nusrat Jahan demo.nusrat@skillswap.local	Mentor	Approved	2026-09-09	Review Delete
Tanvir Rahman demo.tanvir@skillswap.local	Mentor	Approved	2026-09-09	Review Delete
Sadia Islam demo.sadia@skillswap.local	Mentor	Approved	2026-09-09	Review Delete
Arif Hossain demo.arif@skillswap.local	Mentor	Approved	2026-09-09	Review Delete

Skill management



Skillswap monitoring



Admin Store management

The screenshot displays the 'ADMIN STORE' interface for 'SkillSwap'. The top navigation bar includes links for Home, Dashboard, Explore Skills, Store, About, and Contact, along with a user profile icon labeled 'Admin'. The main section is titled 'Store products' and features a 'Dashboard' button and a '+ Add product' button. Below this, there are tabs for 'Products' (4), 'Categories', and 'Orders', with a 'View Public Store' link. A summary section shows: Total products: 4, Active & listed: 4, Out of stock: 0, and Categories: 3. A search bar and filters for 'All categories', 'All stock', and 'All statuses' are present. The main table lists products with columns for Image, Product, Category, Price, Stock, Status, and Actions.

Image	Product	Category	Price	Stock	Status	Actions
	Matte Blue Premium Acoustic Guitar - 2024 Edition ~With 1x Guitar, 5 Picks, Guitar Belt a...	Creative tools	₺3200.00	18	Active	Edit Delete
	A5 Study Notebook A compact notebook for course note...	Learning supplies	₺180.00	21	Active	Edit Delete
	Sketching Pencil Set A practical pencil set for drawing, de...	Creative tools	₺350.00	13	Active	Edit Delete

Admin Add new products

The screenshot shows the 'STORE INVENTORY' section with the title 'Add New Product'. The form includes fields for 'Product Name' (with a placeholder 'e.g. Modern Mechanical Pencil Set'), 'Category' (a dropdown menu), 'Price (₺)' (with a placeholder '0.00'), and 'Initial Stock' (with a placeholder '0'). Below these is a 'Product Description' field with a placeholder 'Describe the product features, specs, and details...' and a note 'Provide clear, helpful details for learners and community buyers.' The 'Product Image' section has a 'Choose File' button and a 'No file chosen' status. At the bottom, it states 'Supported image formats: PNG, JPG, JPEG, WEBP. Maximum file size: 10MB.' and a partially visible 'Active and publicly listed in the Store' button.

Admin Orders Review

Customer orders

Products Categories **Orders 11** View Public Store ↗

All Store Orders (11) Awaiting confirmation orders are highlighted

Order	Customer	Items	Total & Payment	Order Status	Status Management
#15 01 Oct 2026	pratoy barua pratoy16@gmail.com	1 item Matte Blue Premium Acoustic ...	₹3200.00 Paid	Processing Awaiting confirmation	Confirm Order Invoice ↗
#14 23 Sep 2026	pratoy barua pratoy16@gmail.com	1 item Sketching Pencil Set (x4)	₹1400.00 Paid	Delivered	Delivered (Complete) Invoice ↗
#13 23 Sep 2026	pratoy barua pratoy16@gmail.com	1 item A5 Study Notebook (x1)	₹180.00 Paid	Delivered	Delivered (Complete) Invoice ↗
#11 23 Sep 2026	pratoy barua pratoy16@gmail.com	1 item A5 Study Notebook (x1)	₹180.00 Paid	Cancelled	Cancelled Invoice ↗
#10 23 Sep 2026	pratoy barua pratoy16@gmail.com	1 item A5 Study Notebook (x1)	₹180.00 Paid	Delivered	Delivered (Complete) Invoice ↗
#9 23 Sep 2026	pratoy barua pratoy16@gmail.com	1 item A5 Study Notebook (x3)	₹540.00 Paid	Delivered	Delivered (Complete) Invoice ↗
#6 14 Sep 2026	pratoy barua pratoy16@gmail.com	1 item A5 Study Notebook (x2)	₹360.00 Paid	Delivered	Delivered (Complete) Invoice ↗

Notification

LEARNER DASHBOARD

Welcome back, pratoy barua.

Continue your learning journey and connect with skilled mentors.

Active learning: 1 (Current mentor connections)
Pending requests: 0 (Waiting for a response)
Completed learning: 0 (Finished learning)

Active learning
Stay on track with your current learning plans.

Ashik Dash
cyclist - In Progress
CURRENT MILESTONE PROGRESS: 25 of 100%
STATUS: In Progress
EXPECTED FINISH: Sep 26, 2026

Notifications (36 unread)

- Order Placed**
Order #15 is being processed.
Oct 01, 01:47 PM
- Profile Details Unlocked**
You have unlocked Nusrat Jaha...
Oct 01, 01:34 PM
- Profile Details Unlocked**
You have unlocked Nusrat Jaha...
Oct 01, 01:34 PM
- Payment Failed**
Your payment of ₹50.00 to unlo...
Oct 01, 01:34 PM

Mark all as read View all notifications →

My learning
Messages
Update profile

Learning requests View all Recent notifications View all

Mentor approval request

 [Home](#) [Dashboard](#) [Explore Skills](#) [Store](#) [About](#) [Contact](#) [Admin](#)

APPLICATION REVIEW
Mentor Application
Review the mentor's profile, details and verification documents before making a decision.

**Ajoy Roy** MENTOR
ajoy16@gmail.com 01741215122

LOCATION
Dhaka, Mohakhali
Mohakhali DOHS, Block: C, Road: 12, House no:#34

BIO
I'm a passionate web developer with experience in building modern, responsive, and user-friendly web applications. I enjoy working with technologies like HTML, CSS, JavaScript, Python, Flask, and databases. I love helping beginners understand web development through practical projects, problem-solving, and hands-on guidance. Whether you're starting from scratch or looking to improve your development skills, I'm happy to help you learn and grow.

SKILLS
Web Development (4)

TEACHING
4 experience · Online · Paid · ₳3000.00

Private verification documents
These files are protected and only available to authenticated administrators.

Identity
NID_Front.png - 102.7 KB [Download](#)

Certificate
ChatGPT_Image_Sc... 1.48 MB [Download](#)

Protected Documents
Sensitive verification files are stored in a private restricted location and are never exposed publicly. Access is strictly audited and limited to authenticated administrators.

Account Actions
Manage this active account.
[Suspend](#) [Delete](#)

Mentor progress update

 [Home](#) [Dashboard](#) [Explore Skills](#) [Store](#) [About](#) [Contact](#) [Ashik Dash](#)

OVERALL PROGRESS
25%
Status: In Progress

CURRENT FOCUS
Other
Active learning module

TARGET DATE
Sep 26, 2026
Started: Sep 11, 2026

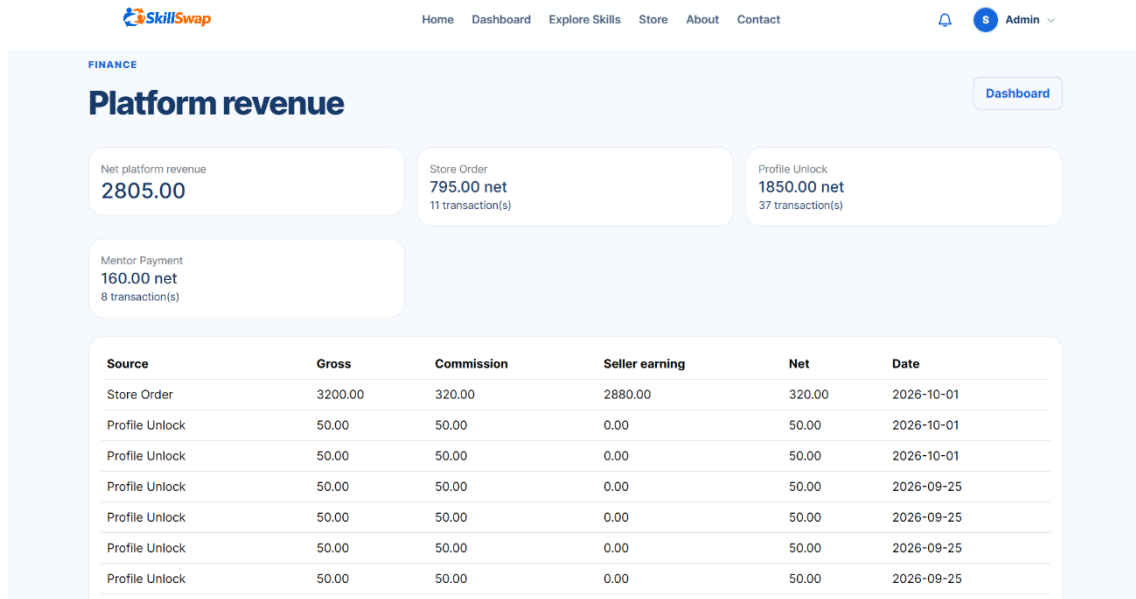
PLAN CADENCE
₳200.00
Weekly

Update Learning Progress [Mentor Control](#)
Progress Percentage
Select the milestone reached in this learning relationship.
25% - Foundations & Basics
0% 25% 50% 75% 100%
Current Topic / Module Focus
Other
The specific topic or concept you and the learner are currently working on.
Completed Topics
balance
Remaining Topics
Enter upcoming topics (one per line or separated by commas)

**Ashik Dash**
Verified Mentor
★ 5.0 [eyelist](#) 3
[Message Mentor](#)
[View Full Profile](#)

Learner Details
**pratoy barua**
Dhaka
[Message Learner](#)

Reports and Analysis



8.3 UI Design Principles

Mandel's Three Golden Rules are foundational principles in User Interface (UI) and User Experience (UX) design, created by interface designer Theo Mandel. Their primary purpose is to keep interfaces intuitive, predictable, and stress-free for the user.

Rule 1: Place the User in Control

Users feel most comfortable when they are driving the action rather than being forced into rigid workflows by software.

- **Allow Undo/Redo:** Users should be able to make mistakes and reverse them easily without penalty.
- **Flexible Navigation:** Do not trap users in a fixed sequence of screens; let them exit or move freely.
- **Accommodate Skill Levels:** Provide shortcuts (like Ctrl+S or macro commands) for power users, while keeping simple step-by-step visual guidance for beginners.

Rule 2: Reduce the User's Memory Load

Human short-term memory is limited. An interface should never require a user to memorize information from one screen to use it on another.

- **Recognition Over Recall:** Display choices visually (e.g., using icons like a trash can or a shopping cart) instead of forcing users to remember specific text commands.
- **Visual Hierarchy:** Group related tools together naturally, and use visual defaults so users don't have to fill out every field from scratch.
- **Maintain Visibility:** Keep vital context, status updates, or instructions visible on-screen during complex tasks.

Rule 3: Make the Interface Consistent

Consistency creates predictability. If a user learns how to perform an action in one part of the software, that knowledge should apply everywhere else.

- **Visual Consistency:** Keep button colors, typography, layout structures, and icon meanings identical across all screens.
- **Behavioral Consistency:** If double-clicking an item opens a detailed view in one module, it should not delete the item in another module.
- **Expectation Matching:** Align the interface with real-world standards (e.g., matching the checkout process of standard e-commerce platforms).

9. Conclusion

SkillSwap provides an integrated web-based platform that connects learners with mentors based on their skills, requirements, and location. The system manages the complete mentor–learner lifecycle, including mentor discovery, learning requests, communication, learning plans, progress tracking, reviews, invoices, payments, and mentor earnings.

The addition of AI-based mentor recommendations, notifications, premium services, an integrated store, and centralized administration makes the platform broader than a basic tutoring marketplace. The system is technically feasible using the selected technologies and can be further expanded with features such as real payment gateways, mobile applications, advanced AI matching, and larger-scale deployment.

10. References

United International University, Department of Computer Science and Engineering. (2026). Software Requirements Specification for Neighborhood Skill Exchange.

1. Github: <https://www.github.com>
2. Superprof- <https://superprof.com/>
3. Preply-- <https://www.preply.com/>
4. MentorHub-- <https://mentorhub.com/>
5. MentorCruise-- <https://mentorcruise.com/>

11. Appendices

Appendix A – Glossary

Learner : User who wants to learn a skill.

Mentor : User who provides knowledge or training.

AI Recommendation : Automated recommendation of suitable mentors.

SRS : Software Requirements Specification document defining system requirements.

DFD : Data Flow Diagram showing how data moves within the system.

UML : Unified Modeling Language used for system design diagrams.

Use Case : Description of user interaction with the system.

Actor : User or entity interacting with the system (Mentors, Learners , Admin).

Dashboard : Interface displaying key system information and features.

IoT : Internet of Things devices used for data collection and monitoring.

Database : Storage system for application data.